SOLUTION
SECTION-I

Single Correct Answer Type **7 Q. [3 M (–1)]**

1. Ans. (B)

Sol. Vibrational energy involves additional degrees of freedom. Molar heat capacity is proportional to the number of degrees of freedom.

2. Ans. (B)

Sol. If gasdoes not escape then volume remains same

$$\frac{P_1}{T_1}=\frac{P_2}{T_2}$$

$$\frac{10^5}{300}=\frac{\left(10^5+\frac{kx}{A}\right)}{T_2};\ \frac{10^5}{300}=\frac{\left(10^5+\frac{1000}{10^{-2}}\right)}{T_2}$$

$$\frac{10^5}{300}=\frac{2\times10^5}{T_2}$$

$T_2 = 600\ K$

$\Delta Q = W + \Delta U = 0 + nC_V\Delta T$

$= 2\times\frac{3R}{2}\times300 = 900\ R$

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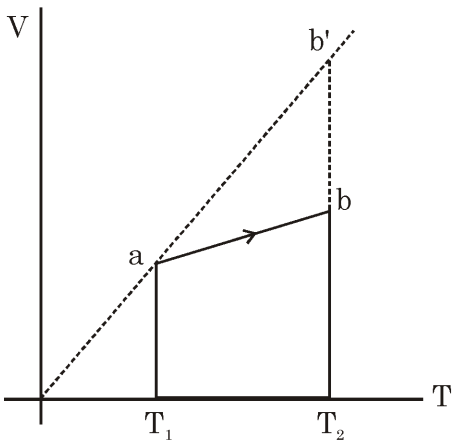
3. Ans. (B)

Sol. $i^2R = nC_P\frac{dT}{dt}+mgv$

$C_P=\frac{7}{2}R$ & $PV = nRT \Rightarrow PAv = nR\frac{dT}{dt}$

4. Ans. (D)

Sol.



If process was ab' then by PV = nRT

P = constant V ∝ T

Now from a → b

Temperature is increasing but product of PV does not increase by equal amount as temperature increases.

Hence no. of moles are decreasing i.e. gas must leak from vessel.

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5. Ans. (B)

Sol. For C', $x = 0 \Rightarrow$ isobaric process $\Rightarrow C' = \frac{5}{2}R$ [Monoatomic gas]

For x', $C = 0 \Rightarrow$ adiabatic process $\Rightarrow x' = \frac{5}{3}$ [Monoatomic gas]

6. Ans. (D)

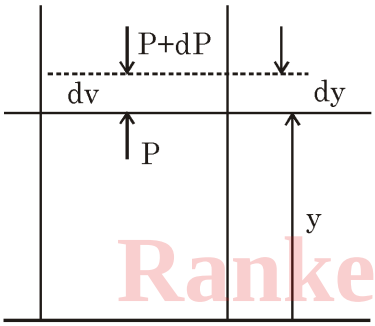
Sol. $\gamma = \frac{dV}{VdT}$

For ideal gas PV = nRT .....(i)
At constant pressure, PdV = nRdT ...(ii)

Dividing (ii) by (i) $\frac{dV}{V} = \frac{dT}{T} \Rightarrow \frac{dV}{VdT} = \frac{1}{T}$

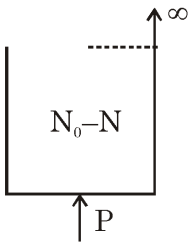
7. Ans. (C)

Sol.



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Let total number of molecules are N_0 and molecules below height y are N.
for volume of gas, from height h to ∞



$P \times A = (N_0 - N)mg$ ... (i)
Where m is mass of 1 molecule
for volume dV,
PdV = dN kT (T is constant)
P × Ady = dN kT
from (i)
$(N_0 - N)$ mg dy = dN kT

$$\frac{dN}{N_0 - N} = \frac{mg}{kT}dy$$

for y = 0, N = 0

$$\int_0^N \frac{dN}{N_0 - N} = \frac{mg}{kT}\int_0^y dy$$

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$$-\left[\ell n\left(N_0 - N\right)\right]_0^N = \frac{mg}{kT}y$$

$$\left(\frac{N_0 - N}{N_0}\right) = e^{-mgy/kT}$$

$$1 - \frac{N}{N_0} = e^{-mgy/kT}$$

$$\frac{N}{N_0} = 1 - e^{-mgy/kT}$$

Multiple Correct Answer Type **6 Q. [4 M (–1)]**

8. Ans. (A, D)

Sol. As, $W_{by} = Q = 0$, So U = const.

$$2 \times \frac{5}{2}R \times T_0 + 3 \times 3R \times 2T_0 = (2 \times \frac{5}{2}R + 3 \times 3R)\, T$$

$$23RT_0 = 14\, RT \Rightarrow T = \frac{23T_0}{14}$$

$$\text{By } \frac{n_1 + n_2}{1 - \gamma_{mix}} = \frac{n_1}{1 - \gamma_1} + \frac{n_2}{1 - \gamma_2} \Rightarrow \gamma_{mix} = \frac{19}{14}$$

9. Ans. (A,B)

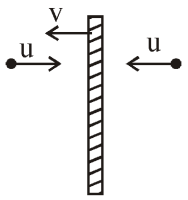
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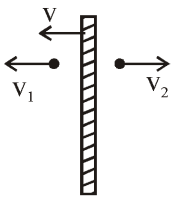
Sol. Total displacement of mercury against atmospheric process= $\left[\ell + \frac{\ell}{2} + \ell\right]$

$PE_{initial} = \left[\left(\rho S \frac{\ell}{2}\right)\frac{\ell}{4}\right] \times 2 + 0$ {Assuming zero at ground level}; $PE_{max}\left[\left(\rho S \frac{\ell}{2}\right) \times 2 + \rho S\ell\right]\ell$

10. Ans. (A,B,D)

Sol.





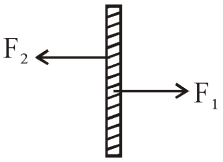
Just before the collision Just after the collision

$v_1 = u + 2v$
$\Delta v_1 = (2u + 2v)$

$v_2 = (u - 2v)$
$\Delta v_2 = (2u - 2v)$

$F_1 = \frac{dp_1}{dt} = \rho A(u + v)(2u + 2v)$
$= 2\rho A (u + v)^2$

$F_2 = \frac{dp_2}{dt} = \rho A(u - v)(2u - 2v)$
$= 2\rho A (u - v)^2$



$\Delta F = F_1 - F_2$ (ΔF net force due to the air molecules on the plate)

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$= 2\rho A\,(4uv) = 8\rho Auv$

$$P = \frac{\Delta F}{A} = 8\rho(uv)$$

$F_{net} = (F - \Delta F) = ma$ (m is mass of the plate)

$F - (8\rho Au)v = ma$ a depends on v

as v is changing, acceleration isnot constant.

(D) After some time

ΔF will increase as v is increasing

$\Delta F = 8\rho Auv$

and finally, $\Delta F = F$

11. Ans. (A,C)

12. Ans. (A, D)

Sol. We know

$$\gamma_{He} = \frac{5}{3}, \gamma_{O_2} = \frac{7}{5}$$

$PV^{\gamma} = \text{const.}$

$$P_{He}\,(LA)^{5/3} = P'_{He}\left(\frac{L}{2}A\right)^{5/3}$$

$$P'_{He} = P_{He}\,(2)^{5/3} \qquad \ldots (1)$$

$$P_{O_2}\,(LA)^{7/5} = P'_{O_2}\,(L'A)^{7/5}$$

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$$P'_{O_2} = P_{O_2}\left(\frac{L}{L'}\right)^{7/5} \qquad \ldots (2)$$

Now $P_{O_2} = P_{He} \Rightarrow P'_{O_2} = P'_{He}$

$$\Rightarrow \frac{P_{O_2}}{P'_{O_2}} = \frac{P_{He}}{P'_{He}} \Rightarrow \left(\frac{L'}{L}\right)^{7/5} = \frac{1}{2^{5/3}}$$

$$(L')^{7/5} = \frac{L^{7/5}}{2^{5/3}}$$

$$L' = \frac{L}{2^{5/3}\,x^{5/7}} = \frac{L}{2^{25/21}}$$

so volume of $O_2 = L'A = \frac{LA}{2^{25/21}}$

13. Ans. (AD)

Sol. For iso-thermal process PV = constant

and for adiabatic process PV^{γ} = constant where $\gamma > 1$

First we assuming that BC is isothermal & CA is adiabatic

$$\Rightarrow P_0V_0 = 3P_0V_C \Rightarrow V_C = \frac{V_0}{3} \text{ [for process BC]}$$

$$3P_0\left(\frac{V_0}{3}\right)^{\gamma} = \frac{V_0P_0}{2} \Rightarrow \gamma = \frac{\ln 6}{\ln 3} \text{ [for process CA]}$$

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Now we are assuming that process
BC is adiabatic & AC is isothermal

$3P_0V_C = \frac{P_0V_0}{2} \Rightarrow V_C = \frac{V_0}{6}$ [for process CA]

$3P_0\left(\frac{V_0}{6}\right)^{\gamma} = P_0V_0^{\gamma} \Rightarrow 6^{\gamma} = 3$ [for process BC] $\Rightarrow V = \frac{\ln 3}{\ln 6}$ here $\gamma < 1$ [not possible]

• Process AC is adiabatic & $V = \frac{\ln 6}{\ln 3}$

Linked Comprehension Type **(1 Para × 4Q.) [3 M (-1)]**
(Single Correct Answer Type)

14. Ans. (B)

Sol. Initially $V_A = V_B = V_0$
$T_A = T_0$, $T_B = T_0$ (always)
$m_A = m$, $P_i = P$

Finally, $V_A = \frac{V_0}{2}$, $V_B = \frac{3}{2}V_0$

$P_A = P_B$ (at equilibrium)
For A
process is adiabatic
$TV^{\gamma-1} = \text{constant}$
$\gamma = 1.5$ (given)
$TV^{1/2} = \text{constant}$

$T_0V_0^{1/2} = T\left(\frac{V_0}{2}\right)^{1/2}$

$T_A = T_0\sqrt{2}$
Now, $P_A = P_B$

$\frac{nRT_A}{V_A} = \frac{nRT_B}{V_B}$

$m\frac{T_0\sqrt{2}}{\frac{V_0}{2}} = m_B\frac{T_0}{\frac{3V_0}{2}}$

$[m_B = 3\sqrt{2}\,m]$

15. Ans. (B)

Sol. $\Delta V = nC_V\Delta T$

$= n \times \frac{R}{\gamma - 1} \times (T_2 - T_1)$

$= \frac{nR(\sqrt{2}-1)T_0}{0.5}$

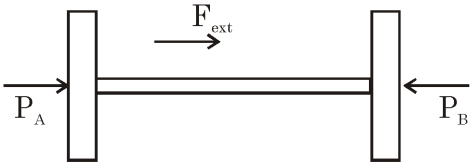
$= PV_0 \times 2(\sqrt{2}-1)$

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16. Ans. (C)

Sol. Since piston is slowly moving towards left



According to work energy theorem.

$W_B + W_{ext} + W_A = 0$

$W_B = -(W_A - W_{ext})$

Since $W_A < 0$, $W_{ext} < 0 \Rightarrow |W_B| = |W_A| + |W_{ext}|$

$|W_B| > |W_A|$

17. Ans. (C)

Sol. At equilibrium,

Compressive force, $F = p_f \times S$

$$\text{For A} \Rightarrow \frac{PV_0}{T_0} = \frac{P_f \times \frac{V_0}{2}}{T_0\sqrt{2}} \Rightarrow P_f = P_x 2\sqrt{2} = 2^{3/2}P$$

SECTION-II

Numerical Answer Type Question (upto second decimal place) **3 Q. [3(0)]**

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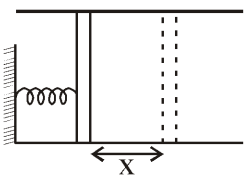
18. Ans. 100

Sol. $P_2V_2 = P_1V_1$

$V_1 = 20000$cc.

$P_1 = 10^5$ Pa

$V_2 = Ax$



$$P_2 = P_0 + \frac{kx}{A}$$

$$\Rightarrow \quad 10^5 \times 20000 \times 10^{-6} = \left(10^5 + \frac{1000x}{100 \times 10^{-4}}\right) 10^{-4} \times 100x$$

$$\Rightarrow \quad 2000 = 1000\,[x\,(1+x)]$$

$$x^2 + x - 2 = 0$$

$$x = \frac{-1 \pm \sqrt{1+8}}{2} = 1 \text{ m}$$

$x = 100$ cm.

19. Ans. 20

Sol. Let initial pressure be P_0

If level A is maintained $\Rightarrow$volume remains same

Temperature double $\Rightarrow$ pressure doubles

$\therefore\ 2P_0 - P_0 = \rho g \times 20$ cm

$\therefore\ P_0$ corresponds to 20 cm of Hg

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20. Ans. 375

Sol. Let us denote the final volume and pressure by $V' = 1.25\ V_0$ and P', respectively. Taking helium to be an ideal gas, we can write that $P_0V_0 = P'V'$ under isothermal conditions, so that $P' = 0.8\ P_0$. Since the external pressure is $0.5\ P_0$, the net outward pressure on the surface of the balloon is $0.3\ P_0$. Thus the net pressure of the balloon is $0.3\ P_0$. Thus the net pressure force on half of the balloon is $0.3\ P\pi R^2$, where R is the final radius of the balloon, given by $1.25\ V = 4\pi R^3/3$. But this must just balance the elastic force $\sigma\Delta A$, where $\Delta A = 2\pi Rt$ is the cross-sectional area of the balloon material at its mid-plane. Here t is the thickness of the balloon material, obtained from $\rho = m/(4\pi R^2 t)$. Combining the equations gives

$$\sigma = \frac{9P_0V_0\rho}{16m}$$

SECTION-III

Numerical Grid Type (Ranging from 0 to 9) 3 Q. [4 M (0)]

21. Ans. 5

Sol. Process A → B

$$W_{AB} = \int P\,dv = \int \frac{3}{2}T^{1/2}\,dv = \int \frac{3}{2}T^{1/2} \times \frac{1}{3}RT^{-1/2}\,dT$$

On solving, $W_{AB} = \frac{115.2}{2} \times R = 480\ J$

Process B → C

$$U = \frac{1}{2}V^{1/2} \quad \frac{3}{2}RT = \frac{1}{2}V^{1/2} \Rightarrow 3PV^{1/2} = 1$$

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$$\therefore P = \frac{1}{3\sqrt{V}}$$

$$\text{New } W_{BC} = \int P\,dv = \int_{100}^{1600} \frac{1}{3\sqrt{V}}\,dv = \frac{2}{3}\sqrt{V} = \frac{2}{3}[40 - 10] = \frac{2}{3} \times 30 = 20\ J$$

Total work $W = 480 + 20 = 500\ J$

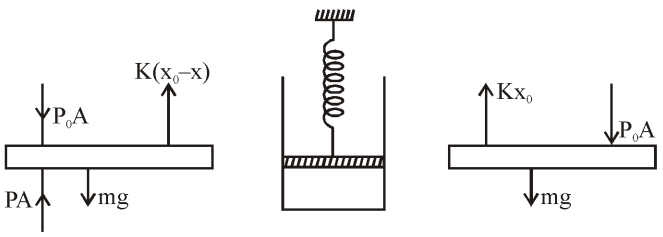
22. Ans. 8

Sol. $P_0A + mg = kx_0$

for very slow process

$PA = mg + p_0A - K(x_0 - x)$

$$\Rightarrow P = \frac{Kx}{A} = \frac{KV}{A^2}$$



Thus process followed by gas is $PV^{-1} = \frac{K}{A^2} = \text{constant}$.

$\therefore$ Molar heat capcity for process is

$$= C = C_v + \frac{R}{2} = 2R \quad \Rightarrow Q = n.C\Delta T$$

$$\text{Since } \frac{P}{V} = \frac{AK}{A^2} \Rightarrow T = \frac{KV^2}{nRA^2}$$

$$\Rightarrow \Delta T = \frac{K}{nR}\left(1 - \frac{1}{9}\right)\ell^2 \Rightarrow Q = \frac{16K}{9}\ell^2$$

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23. Ans. 3

Sol. We have

$\Delta Q = \Delta U + \Delta W$

At constant pressure,

$\Delta Q = C_P \Delta T$

$\Delta U = C_V \Delta T$

$$\Delta W = P\Delta V = PV_0 (3\alpha)\, \Delta T = \frac{PM}{\rho} (3\alpha)\, \Delta T$$

$$\therefore\ C_P = C_V + \frac{PM}{\rho} (3\alpha)$$

$$\Rightarrow C_P - C_V = 3\alpha\left(\frac{MP}{\rho}\right)$$

SECTION-IV

Matrix Match Type (4 × 5) **2 Q. [8 M (for each entry +2(0)]**

24. Ans. (A) P (B) Q (C) Q

Sol. For (A)

(A) For cyclic process ABCA

$Q_{ABC\ circle} + Q_{CA\ direct} > 0$

Since cycle is clockwise

Now $Q_{CA\ Direct} = 0$, process is adiabatic so

$\Delta Q_{ABC\ circle} > 0$

$A \rightarrow C$ from circle $\Delta Q > 0$

For (B)

$B \rightarrow C$

$W < 0$ (compresion)

$\Delta U < 0$ (PV product decreases) $\Rightarrow \Delta Q < 0$

For (C)

$C \rightarrow D$

$W < 0$ (compresion)

$\Delta U < 0$ (PV product decreases) $\Rightarrow \Delta Q < 0$

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25. Ans. (A) R; (B) P,Q; (C) P,Q; (D) S,T

Sol. For process-1,

Gas undergoes compression work done by gas is –ve

for DU & DQ nothing can be comented.

For process-2,

Gas undergoes expansion Þ work done by gas is +ve

Also temperature of gas increases.

$\therefore$ ΔQ is +ve & ΔU is +ve

For process-3

Process is isobaric

For process-4

Process is iso-choric

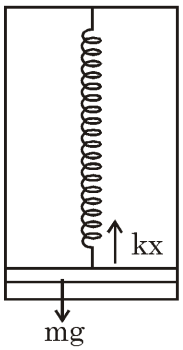
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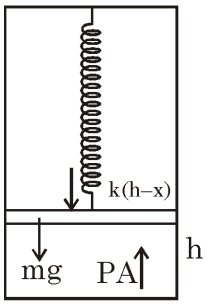
5 Q. [4 M (0)]

26. **Ans.** $h' = \sqrt{2}\,h$

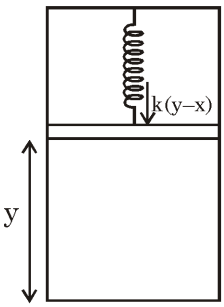
Sol.



$kx = mg$



$PA = mg + k(h - x)$
$PA = kh$
$V = Ah$
T



$P'A = mg + k(y - x)$
$P'A = kg$
$V' = Ay$
$2T$

$$\frac{PV}{T} = \frac{P'V'}{2T}$$

$$\frac{\frac{kh}{A} \times Ah}{T} = \frac{\frac{ky}{A} \times Ay}{2T}$$

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$y^2 = 2h^2$

$y = \sqrt{2}\,h$

27. **Ans. 0, 0, 14650/3 cal**

Sol. The cylinder is perfectly conducting and heat has been provided slowly therefore in the absence of any temperature gradient entire heat given to gas is absorbed by ice. Phase changes take place at constant temperature and pressure.

(a) $W_{gas} = 0$ (b) $Q_{gas} = 0$

(c) Heat absorbed by ice water system to reach 100°C

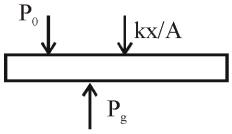
$= mL + 2ms_w(100) = 10 \times 80 + 2 \times 10 \times 1 \times 100 = 2800$ Cal

$$= nC_p\Delta T = 3 \times \frac{7}{2}R \times (100) = 3 \times \frac{7}{2} \times \frac{25}{3} \times \frac{1}{42} \times 100 = 6250/3 \text{ Cal}$$

Total heat = 2800 + 6250/3 = 14650/3 Cal

28. **Ans.** 1.6 m, 364 K

Sol.



$$P_g = P_0 + \frac{kx}{A} = P_0 + \frac{kh}{16A}$$

For adiabatic process $P_1V_1^{\gamma} = P_2V_2^{\gamma}$

$$P_0(hA)^{\gamma} = P_g\left[\left(\frac{h}{2} + \frac{h}{16}\right)A\right]^{\gamma}$$

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$$10^5 (hA)^{\gamma} = \left[10^5 + \frac{3700 \times h}{16 \times 27 \times 10^{-4}}\right]\left[\frac{9h}{16} A\right]^{\gamma}$$

$$\frac{64}{27} = 1 + \frac{370 \times h}{16 \times 27}$$

$$\frac{37}{27} = \frac{370 \times h}{16 \times 27} \Rightarrow h = \frac{16}{10}$$

PV = nRT

P_0hA =nR × 273 (initial)

$$\left\{P_0 + \frac{kh}{16A}\right\}\left\{\frac{Ph}{16} A\right\} = nRT \quad \text{(final)}$$

divide

$$\left\{1 + \frac{kh}{16A \times P_0}\right\}\frac{9}{16} = \frac{T}{273}$$

$$T = 273 \times \frac{9}{16}\left[1 + \frac{370}{270}\right]$$

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$$T = 273 \times \frac{4}{3} = 364k$$

29. **Ans.** C = 2R

Sol. $C = C_V + \frac{PdV}{dT}$

Now

PdV + VdP = nRdT

$$PdV = \frac{k\ell}{A} Adx = k\ell dx$$

$$\left(\frac{k\ell}{A}\right)(Adx) + (A\ell)\frac{kdx}{A} = 1 \times RdT$$

$$\frac{2k\ell dx}{dT} = R$$

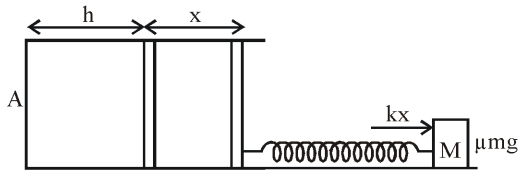
$$\frac{k\ell dx}{dT} = \frac{R}{2}$$

$$C = \frac{3}{2}R + \frac{R}{2} = 2R$$

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30. **Ans.** (a) 2.515, (b) 404.8 K, (c) 122.179 J

Sol.



(a) $\mu mg = 3 = kx \Rightarrow x = \frac{3}{k} = 10^{-2}$

$$W = P_0 A \times x + \frac{1}{2}kx^2 = 10^5 \times 25 \times 10^{-4} \times 10^{-2} + \frac{1}{2} \times 300 \times 10^{-4}$$

$= 2.5 + 0.015 = 2.515$ J

(b) $P_0 = 10^5$ N/m^2, $T_0 = 400$ K, $V_0 = Ah$

$$P_2 = P_0 + \frac{kx}{A} = 1.012 \times 10^5 \text{ N/m}^2$$

$V_2 = V_0 + Ax$

Initial $P_0 \times (Ah) = nR \times 400$

final $A\left(P_0 + \frac{kx}{A}\right)(h + x) = nR \times T_f$

$\Rightarrow T_f = 404.8$

(c) $\Delta Q = W + nC_V\Delta T$

$$= W + \frac{2 \times 3}{2} R\left[T_f - 400\right]$$

$= 2.515 + 3 \times 8.31 \times 4.8$

$= 2.515 + 119.664$

$= 122.179$

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